Increased Prevalence and Mortality in Undiagnosed Celiac Disease

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Background & Aims The historical prevalence and long-term outcome of undiagnosed celiac disease (CD) are unknown. We investigated the long-term outcome of undiagnosed CD and whether the prevalence of undiagnosed CD has changed during the past 50 years. Methods This study included 9,133 healthy young adults at Warren Air Force Base (sera were collected between 1948 and 1954) and 12,768 sex-matched subjects from 2 recent cohorts from Olmsted County, Minnesota, with either similar years of birth (n=5,558) or age at sampling (n=7,210) to that of the Air Force cohort. Sera were tested for tissue transglutaminase and, if abnormal, for endomysial antibodies. Survival was measured during a follow-up period of 45 years in the Air Force cohort. The prevalence of undiagnosed CD between the Air Force cohort and recent cohorts was compared. Results Of 9,133 persons from the Air Force cohort, 14 (0.2%) had undiagnosed CD. In this cohort, during 45 years of follow-up, all-cause mortality was greater in persons with undiagnosed CD than among those who were seronegative (hazard ratio=3.9; 95% CI, 2.0–7.5; $P<.001$). Undiagnosed CD was found in 68 (0.9%) persons with similar age at sampling and 46 (0.8%) persons with similar years of birth. The rate of undiagnosed CD was 4.5-fold and 4-fold greater in the recent cohorts (respectively) than in the Air Force cohort (both $P \leq .0001$). Conclusions During 45 years of follow-up, undiagnosed CD was associated with a nearly 4-fold increased risk of death. The prevalence of undiagnosed CD appears to have increased dramatically in the United States during the past 50 years.